

Contrasting Seasonal Rainfall Trends under Predominantly Stable Annual Conditions: Evidence from Trend-Free Pre-Whitening Analysis in Prachuap Khiri Khan Province, Thailand

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Contrasting Seasonal Rainfall Trends under Predominantly Stable Annual Conditions: Evidence from Trend-Free Pre-Whitening Analysis in Prachuap Khiri Khan Province, Thailand

Abstract

Statistically stable annual rainfall can mask substantial shifts in seasonal rainfall distribution, obscuring hydroclimatically important changes in rainfall timing and seasonal water availability. This study investigated annual, wet-season, and dry-season rainfall trends at 12 rain-gauge stations in Prachuap Khiri Khan Province, coastal western Thailand, using daily rainfall records from 1981–2014. Serial autocorrelation was evaluated prior to trend detection, and rainfall trends were analysed using a conditional Trend-Free Pre-Whitening Mann–Kendall (TFPW-MK) framework with Sen's slope and rank-based 95% confidence intervals. Among 36 station–timescale combinations, seven (19.4%) exhibited statistically significant trends. Annual rainfall remained predominantly stable, with a significant increase detected only at station 500002 (+9.18 mm yr⁻¹). In contrast, significant wet-season declines occurred at three stations (–5.61 to –7.62 mm yr⁻¹), whereas three stations showed significant dry-season increases (+5.22 to +6.78 mm yr⁻¹). The strongest seasonal contrast occurred at station 500006, where significant wet-season drying and dry-season wetting coexisted despite the absence of a significant annual trend, providing evidence consistent with intra-annual rainfall redistribution. Accounting for serial dependence also altered trend inference for wet-season rainfall at station 500003, where TFPW-MK identified a significant decline that was not detected by the original Mann–Kendall

test. These results demonstrate that seasonal rainfall redistribution, rather than changes in annual rainfall totals, represents the dominant expression of recent rainfall variability in the study area. Annual rainfall alone therefore provides an incomplete basis for assessing hydroclimatic change, underscoring the importance of season-specific rainfall analysis for water-resource management and climate adaptation in monsoon-influenced coastal regions.

Keywords: Rainfall trend; Seasonal rainfall; Trend-Free Pre-Whitening; Mann–Kendall test; Sen’s slope; Serial correlation; Prachuap Khiri Khan

1. Introduction

Reliable seasonal rainfall underpins water security, agricultural production, ecosystem functioning, and climate adaptation in monsoon regions. However, rainfall change is not always expressed as a simple increase or decrease in annual rainfall totals (Trenberth et al, 2003). A location may exhibit statistically stable annual rainfall while undergoing substantial changes in the seasonal distribution, timing, or persistence of precipitation (Palizdan et al, 2015). Such changes can alter reservoir recharge, soil-moisture replenishment, irrigation scheduling, and seasonal water availability despite little apparent change in annual rainfall (Ahmad et al, 2015). Consequently, analyses based solely on annual rainfall totals may overlook hydroclimatically important changes that are directly relevant to environmental management and water-resource planning (IPCC, 2023).

Thailand's rainfall is governed by the interaction of the Southwest and Northeast monsoons, tropical weather disturbances, regional moisture transport, and complex topography, resulting in pronounced spatial and seasonal variability, particularly in coastal regions (Arai et al, 2019). Previous studies have reported substantial geographical and seasonal heterogeneity in rainfall across Thailand and Southeast Asia (Limsakul et al., 2017; Fiaz et al., 2025), including changes in rainfall concentration, seasonal timing, rainy-season duration, and contrasting annual and seasonal trends. These findings suggest that annual rainfall totals alone may not adequately capture ongoing hydroclimatic change and emphasize the importance of examining seasonal rainfall behaviour to improve understanding of environmental change in monsoon-influenced regions.

Prachuap Khiri Khan Province is well suited for investigating seasonal rainfall variability because its coastal setting and diverse physiography create pronounced spatial differences in rainfall over relatively short distances (Harmoko et al., 2026). Such heterogeneity provides an opportunity to

examine how seasonal rainfall behaviour varies among stations within a relatively compact monsoon-influenced region. Improved understanding of these variations is important for supporting water-resource management, agriculture, and climate adaptation in coastal western Thailand.

Although rainfall variability in Thailand has been widely investigated at national and regional scales, relatively few studies have examined whether statistically stable annual rainfall conceals contrasting seasonal changes at individual monitoring stations in coastal western Thailand. Most previous studies have focused on annual rainfall trends or regional averages, which may mask localized seasonal responses (Deng et al., 2025). Consequently, it remains unclear whether apparent annual stability reflects genuine hydroclimatic stability or results from opposing wet- and dry-season trends that offset one another. Addressing this knowledge gap is important because changes in seasonal rainfall distribution may substantially affect water availability and environmental management without producing detectable changes in annual rainfall totals.

Another important consideration is the influence of serial autocorrelation on trend detection (Burn and Elnur, 2002). The Mann–Kendall (MK) test is widely used for detecting monotonic trends in hydroclimatic time series because it is robust and requires no assumption of normality (Mann, 1945; Kendall, 1975). However, the test assumes serially independent observations, and positive autocorrelation may affect statistical inference by altering the variance of the test statistic (Hamed and Rao, 1998). Trend-Free Pre-Whitening (TFPW) has been widely adopted to reduce the influence of lag-1 serial dependence while preserving the underlying monotonic trend (Yue et al., 2002). Because serial autocorrelation is not present in every rainfall series, applying pre-whitening indiscriminately may introduce unnecessary data transformation, whereas ignoring significant persistence may reduce the reliability of trend detection. Therefore, this study adopted a conditional TFPW framework in which serial autocorrelation was evaluated before trend analysis, and pre-whitening was applied only to rainfall series exhibiting significant lag-1 autocorrelation (Do, 2026).

Against this background, this study examined annual, wet-season, and dry-season rainfall trends using daily rainfall records from 12 rain-gauge stations in Prachuap Khiri Khan Province during 1981–2014. The objectives were to (1) determine whether annual rainfall exhibited significant monotonic trends, (2) examine whether seasonal analyses revealed contrasting wet- and dry-season trends that were not evident from annual rainfall totals, and (3) assess whether accounting for significant serial autocorrelation influenced trend detection relative to the original Mann–Kendall (MK) test. By integrating seasonal rainfall analysis with a conditional Trend-Free Pre-Whitening Mann–Kendall (TFPW-MK) framework, Sen's slope estimation, and station-scale spatial analysis, this study

investigated whether apparently stable annual rainfall concealed environmentally important seasonal changes. The findings provide a more comprehensive understanding of rainfall variability in coastal western Thailand and offer information to support water-resource management, agricultural planning, and climate-change adaptation in monsoon-influenced environments.

2. Study Area and Methods

2.1 Study area

Prachuap Khiri Khan Province is located along the western coast of the Gulf of Thailand in upper southern Thailand and covers approximately 6,368 km². The province extends in a narrow north–south direction, bordered by the Gulf of Thailand to the east and the Tenasserim Range along the Myanmar border to the west. This distinctive physiographic setting creates pronounced topographic gradients over relatively short distances. These gradients strongly influence local rainfall through interactions among the southwest monsoon, coastal moisture transport, and orographic uplift.

The study area encompasses four major physiographic settings: coastal lowlands bordering the Gulf of Thailand, inland alluvial plains, upland and foothill areas associated with the Tenasserim mountain system, and offshore islands in the Gulf of Thailand. Together, these contrasting environments produce substantial variations in elevation, distance from the coastline, and exposure to prevailing monsoon airflow, allowing nearby stations to experience markedly different rainfall regimes.

2.2 Rainfall data and temporal aggregation

Daily rainfall observations from twelve rain-gauge stations located across Prachuap Khiri Khan Province were obtained for the period from 1 January 1981 to 31 December 2014 (Table 1; Figure 1). The station network provides representative coverage of the province's four principal physiographic environments—coastal lowlands, inland alluvial plains, upland areas, and offshore islands—allowing rainfall variability to be evaluated across contrasting geographical settings.

Before analysis, the rainfall records were screened for completeness and temporal continuity to ensure consistent time-series construction. Daily rainfall was then aggregated into three temporal scales: annual (January–December), wet season (May–October), and dry season (November–April). The annual and wet-season datasets each comprised 34 observations spanning 1981–2014, exceeding the 30-year minimum recommended by the World Meteorological Organization for climate normals and long-term climate analyses (WMO, 2017). Because the dry season extends across two calendar years, rainfall totals were accumulated from November of one year through April of the next. As observations

began on 1 January 1981, the first complete dry season could not be constructed, yielding a dataset of 33 complete November–April seasons.

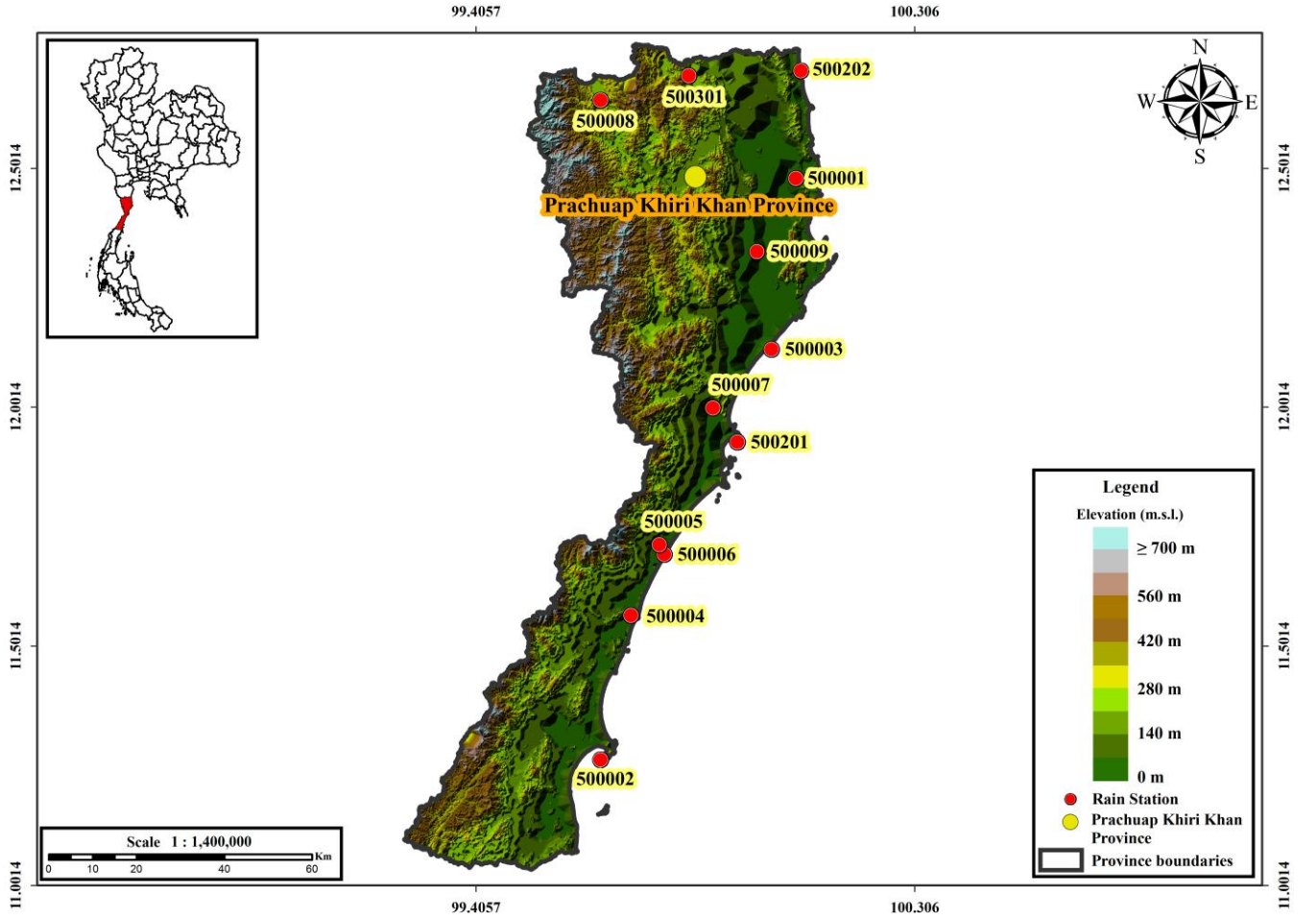


Figure 1. Location of Prachuap Khiri Khan Province and the 12 rainfall stations used in the analysis.

2.3 Serial correlation screening

Serial correlation was evaluated prior to trend analysis because temporal dependence can influence the statistical inference of the Mann–Kendall (MK) test, which assumes serially independent observations. Accordingly, annual, wet-season, and dry-season rainfall series were screened for lag-1 autocorrelation before trend detection.

The lag-1 autocorrelation coefficient (r_1) was calculated as:

$$r_1 = \frac{\sum_{t=1}^{n-1} (x_t - \bar{x})(x_{t+1} - \bar{x})}{\sum_{t=1}^n (x_t - \bar{x})^2} \quad (1)$$

where x_t denotes the annual or seasonal rainfall total at time t , $\bar{x}$ is the mean of the time series, and n is the number of observations.

The statistical significance of r_1 was evaluated using the approximate 95% confidence limits under the null hypothesis of serial independence:

$$UB, LB = \frac{-1 \pm 1.96\sqrt{n-2}}{n-1} \quad (2)$$

For $n = 33-34$, the confidence limits from Eq. (2) are very similar to the widely used symmetric approximation $\pm \frac{1.96}{\sqrt{n}}$. The symmetric approximation was therefore used to identify significant lag-1 autocorrelation.

Rainfall series with autocorrelation coefficients outside these confidence limits were classified as exhibiting significant lag-1 serial correlation. These series were subsequently analysed using the conditional Trend-Free Pre-Whitening Mann–Kendall (TFPW-MK) procedure described in Section 2.4, whereas series without significant autocorrelation were analysed directly using the original MK test. Thus, the serial-correlation screening determined whether persistence correction was required before trend detection.

2.4 Trend-Free Pre-Whitening Mann–Kendall procedure

The Mann–Kendall (MK) test is widely used to detect monotonic trends in hydroclimatic series because it requires no assumption of normality. However, serial correlation can affect statistical inference by altering the variance of the test statistic. To avoid unnecessary correction, pre-whitening was applied only to the seven station–timescale series that exhibited significant lag-1 autocorrelation (Section 2.3), while the remaining series were analysed using the original MK test (Yue et al., 2002).

For each rainfall series exhibiting significant lag-1 autocorrelation, the TFPW procedure comprised three sequential steps. First, Sen's slope (β) was estimated and removed from the original series.

$$y_t = x_t - \beta t \quad (3)$$

Second, the lag-1 autocorrelation coefficient (r_1) was estimated from the detrended series and used to remove residual serial dependence:

$$y_t' = y_t - r_1 y_{t-1} \quad (4)$$

Finally, the estimated trend was restored to the pre-whitened series to produce the trend-preserved series used for the subsequent Mann–Kendall analysis:

$$x_t' = y_t' + \beta t \quad (5)$$

The Mann–Kendall test was subsequently applied to the trend-restored series (x_t'). Rainfall series without significant lag-1 autocorrelation were analysed directly using the original MK test. Because lag-1 pre-whitening requires a preceding observation, each pre-whitened series contained one fewer observation than the original series, resulting in an effective sample size of $n_{\text{eff}} = 33$ for annual and wet-season datasets subjected to pre-whitening. Series that did not require pre-whitening retained the original sample size ($n = 34$). All subsequent trend statistics, including the Mann–Kendall test statistic, Sen's slope, and its rank-based 95% confidence interval, were calculated from the same trend-restored pre-whitened series to ensure methodological consistency between trend detection and trend magnitude estimation within the conditional TFPW-MK framework.

The Mann–Kendall test statistic (S) is determined by summing the signs of all pairwise differences across the time series:

$$S = \sum_{i=1}^{n-1} \sum_{j=i+1}^n \text{sgn}(x_j - x_i) \quad (6)$$

where

$$\text{sgn}(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases} \quad (7)$$

The variance of S , corrected for tied ranks, is given by

$$\text{Var}(S) = \frac{1}{18} \left[n(n-1)(2n+5) - \sum_{p=1}^g t_p(t_p-1)(2t_p+5) \right] \quad (8)$$

where g is the number of tied groups and t_p is the number of observations in the p -th tied group.

The standardized test statistic (Z) was calculated as:

$$Z = \begin{cases} \frac{S-1}{\sqrt{\text{Var}(S)}} & \text{if } S > 0 \\ 0 & \text{if } S = 0 \\ \frac{S+1}{\sqrt{\text{Var}(S)}} & \text{if } S < 0 \end{cases} \quad (9)$$

Positive values of Z indicate increasing trends, whereas negative values indicate decreasing trends. Statistical significance was assessed using a two-sided test at $\alpha = 0.05$.

2.5 Trend magnitude and uncertainty

Trend magnitude was quantified using Sen's slope estimator (Sen, 1968), a nonparametric method that estimates the median rate of change and is robust to outliers. For a time series of length n , all possible pairwise slopes were calculated as

$$Q_k = \frac{x_j - x_i}{j - i}, \forall j > i \quad (10)$$

where x_i and x_j are rainfall totals at times i and j , respectively.

Sen's slope (β) was then estimated as the median of all pairwise slopes:

$$\beta = \text{median}(Q_k) \quad (11)$$

Positive values of β indicate increasing rainfall trends, whereas negative values indicate decreasing rainfall trends. Trend magnitudes are reported in millimetres per year (mm yr^{-1}).

Rank-based 95% confidence intervals for Sen's slope were calculated following the nonparametric procedure described by Sen (1968) and Gilbert (1987). For rainfall series subjected to pre-whitening, both Sen's slope and its confidence intervals were estimated from the same trend-restored pre-whitened series used in the final TFPW-MK analysis to ensure methodological consistency.

2.6 Spatial representation of rainfall trends

Spatial patterns of rainfall trends were mapped using station-based Sen's slope estimates derived from annual and seasonal rainfall series. Continuous trend surfaces were generated by inverse distance weighting (IDW; power = 2), clipped to the administrative boundary of Prachuap Khiri Khan Province, and overlaid with station locations indicating trend direction and statistical significance ($\alpha = 0.05$).

IDW is a deterministic interpolation method and was used only to visualize broad spatial patterns in rainfall trends. Because the interpolated surfaces are influenced by station density and distribution, particularly within the province's sparse monitoring network and heterogeneous coastal terrain, they should not be interpreted as statistically reliable estimates for unsampled locations. Statistical significance was assessed exclusively from the station-level TFPW-MK results, whereas the interpolated maps served only as visual support for regional pattern interpretation. All analyses,

including serial-correlation screening, conditional TFPW–MK, Sen's slope estimation with rank-based confidence intervals, and IDW mapping, were performed in Python 3.12 using NumPy, SciPy, and pandas for statistical analyses, and Matplotlib and GeoPandas for spatial visualization.

3. Results and Discussion

3.1 Rainfall variability and serial dependence

Substantial spatial differences in rainfall characteristics were observed across the 12 monitoring stations, reflecting the heterogeneous hydroclimatic setting of Prachuap Khiri Khan Province (Table 1). Mean annual rainfall ranged from 944.1 mm yr⁻¹ at station 500202 to 1,376.2 mm yr⁻¹ at station 500002, whereas the coefficient of variation (CV) ranged from 9.6% at station 500008 to 27.1% at station 500001. Stations with similar mean annual rainfall did not necessarily exhibit comparable interannual variability. For example, stations 500008 and 500201 received similar annual rainfall totals (1,206.3 and 1,108.2 mm yr⁻¹, respectively), but their variability differed markedly (CV = 9.6% and 24.5%, respectively). These results indicate considerable spatial heterogeneity in rainfall variability across the province.

Serial dependence also varied among stations and temporal scales. Of the 36 station–timescale series analysed, seven (19.4%) exhibited significant lag-1 autocorrelation and therefore required conditional Trend-Free Pre-Whitening (TFPW) prior to trend analysis. All autocorrelated series occurred in the annual or wet-season datasets, whereas none of the dry-season series showed significant lag-1 autocorrelation. This pattern indicates that rainfall persistence differed among seasonal timescales rather than representing a common characteristic of all rainfall records (Endo et al., 2009; Dinpashoh et al., 2014).

The greater persistence observed in the annual and wet-season series may reflect the cumulative influence of interannual climate variability on seasonal rainfall accumulation. However, because atmospheric circulation was not examined directly, this interpretation should be regarded as a plausible physical explanation rather than direct evidence of climate forcing. The observed heterogeneity in autocorrelation supports the conditional application of TFPW, ensuring that pre-whitening was applied only to rainfall series exhibiting significant serial dependence before trend detection (Blain, 2014).

Table 1

Locations and descriptive statistics of annual rainfall at the 12 stations in Prachuap Khiri Khan Province during 1981–2014.

Station	Latitude (°N)	Longitude (°E)	Mean annual rainfall (mm yr ⁻¹)	CV (%)
500001	12.37	99.95	983.70	27.10
500002	11.18	99.55	1376.20	16.10
500003	12.02	99.90	1100.00	19.80
500004	11.47	99.65	1106.00	20.00
500005	11.60	99.68	1147.20	20.40
500006	11.62	99.67	1203.10	13.40
500007	11.90	99.78	1133.60	15.90
500008	12.53	99.55	1206.30	9.60
500009	12.22	99.87	1176.50	9.70
500201	11.83	99.83	1108.20	24.50
500202	12.59	99.96	944.10	23.70
500301	12.58	99.73	1064.80	22.30

3.2 Overall pattern of annual and seasonal rainfall trends

The conditional TFPW-MK analysis showed that rainfall change during 1981–2014 was expressed more clearly at the seasonal than at the annual timescale. Of the 36 station–timescale series analysed, seven (19.4%) exhibited statistically significant monotonic trends at the 5% significance level, comprising one annual increase, three wet-season decreases, and three dry-season increases. The remaining 29 series showed no significant monotonic trends, although both positive and negative Sen's slope estimates were observed (Figure 2).

Seasonal analyses revealed a more coherent pattern than annual rainfall alone. Station 500006 showed a significant decrease in wet-season rainfall together with a significant increase in dry-season rainfall, while annual rainfall remained statistically unchanged. This contrasting seasonal behaviour indicates that annual rainfall stability can conceal important changes in the seasonal distribution of rainfall. Consistent with this finding, all significant seasonal trends occurred as either wet-season decreases or dry-season increases, with no significant wet-season increases detected.

These results suggest that rainfall change in Prachuap Khiri Khan was characterised primarily by shifts in seasonal rainfall distribution rather than by widespread changes in annual rainfall totals. This pattern should not be interpreted as a direct transfer of rainfall between seasons but instead as increasing differentiation in rainfall occurrence throughout the year while annual accumulation remained relatively stable.

From a water-resources perspective, seasonal rainfall changes may have greater practical importance than changes in annual totals because reservoir recharge, irrigation scheduling, soil-moisture replenishment, and water availability depend strongly on the timing of rainfall (Asfaw et al., 2018). These findings demonstrate that seasonal analysis provides a more informative assessment of hydroclimatic change than annual rainfall totals alone in coastal western Thailand.

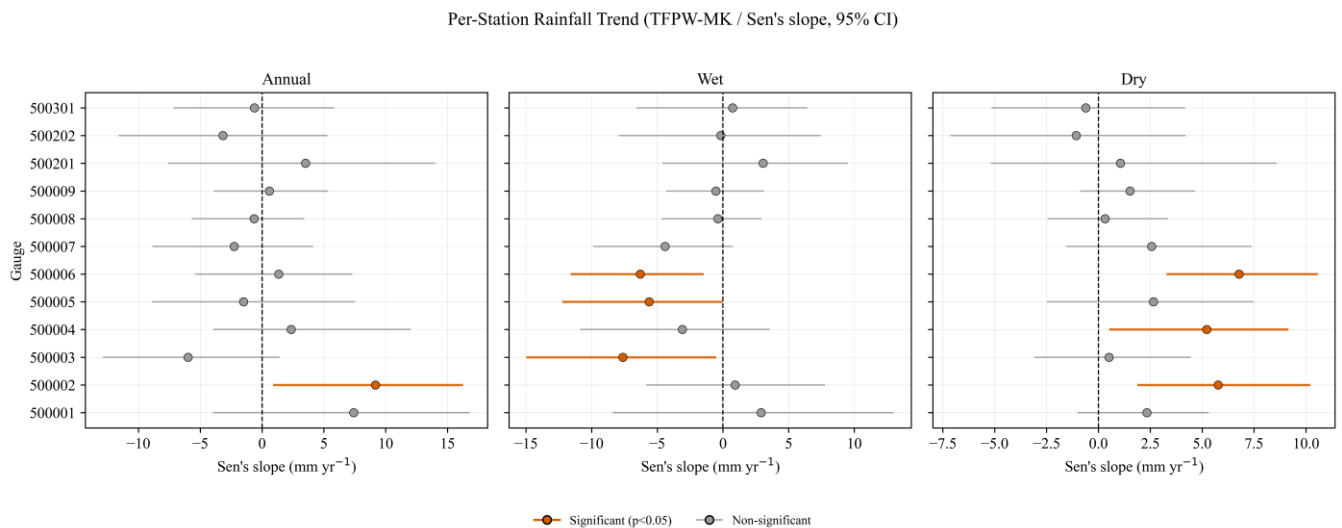


Figure 2. Station-specific annual and seasonal rainfall trends based on Sen's slope estimates and rank-based 95% confidence intervals (1981–2014).

3.3 Annual rainfall trends

Annual rainfall remained largely stable across Prachuap Khiri Khan Province during 1981–2014 (Table 2; Figure 2). Of the 12 monitoring stations, only station 500002 exhibited a statistically significant increasing trend, with a Sen's slope of $+9.18 \text{ mm yr}^{-1}$ (95% CI: $+0.99$ to $+16.15 \text{ mm yr}^{-1}$; $p = 0.024$). The remaining stations showed both positive and negative Sen's slope estimates, but none were statistically significant because their 95% confidence intervals included zero.

The predominance of non-significant annual trends suggests that total annual rainfall changed little over the study period. However, annual rainfall represents the cumulative effect of precipitation across different seasons and may therefore mask contrasting seasonal changes occurring within the same year. Consequently, annual rainfall alone provides limited information for identifying changes in rainfall timing and seasonal distribution (Pascale et al., 2015).

The spatial variability of annual Sen's slope estimates indicates that rainfall responses differed among monitoring stations, reflecting the heterogeneous hydroclimatic characteristics of the province. Although annual rainfall appeared generally stable, the seasonal analyses presented in the following sections reveal changes that are not evident from annual rainfall totals alone.

Table 2.

Annual rainfall trends based on the TFPW-MK framework and Sen's slope estimates.

Station	n_{eff}	τ	Z	p	Sen's slope (mm yr ⁻¹)	CI low	CI high	r_i
500001	34	0.139	1.142	0.254	7.41	-3.92	16.77	0.229
500002	34	0.273	2.253	0.024	9.18	0.99	16.15	-0.139
500003	33	-0.182	-1.472	0.141	-5.99	-12.81	1.36	0.455
500004	34	0.066	0.534	0.594	2.36	-3.89	11.97	0.189
500005	33	-0.023	-0.170	0.865	-1.50	-8.84	7.48	0.405
500006	34	0.066	0.534	0.594	1.35	-5.36	7.24	0.172
500007	34	-0.102	-0.830	0.406	-2.25	-8.81	4.07	0.153
500008	34	-0.034	-0.267	0.790	-0.65	-5.63	3.36	0.018
500009	34	0.057	0.460	0.646	0.58	-3.85	5.30	0.170
500201	34	0.077	0.623	0.534	3.51	-7.52	13.96	0.054
500202	33	-0.095	-0.759	0.448	-3.17	-11.55	5.24	-0.362
500301	34	-0.037	-0.297	0.767	-0.62	-7.10	5.77	-0.105

Note: n_{eff} denotes the effective sample size used in the final MK test. Pre-whitened series had $n_{eff} = 33$ because one observation was required for the lag-1 transformation; all other series retained $n_{eff} = 34$.

3.4 Wet-season rainfall trends

Wet-season rainfall exhibited clearer evidence of change than annual rainfall (Table 3; Figure 2). Significant decreasing trends were detected at stations 500003, 500005, and 500006, with Sen's slopes of -7.62 , -5.61 , and -6.29 mm yr⁻¹, respectively (95% confidence intervals excluding zero). All three series showed significant lag-1 autocorrelation and were therefore analysed using the conditional TFPW-MK procedure.

The significant declines were confined to three stations, whereas the remaining stations exhibited weak positive or negative trends that were not statistically significant. This spatial variability indicates that changes in wet-season rainfall were localized rather than uniform across the province. Such heterogeneity is consistent with the diverse physiographic characteristics of Prachuap Khiri Khan, where coastal proximity and the Tenasserim Range may influence local rainfall patterns over relatively short distances (Khalil et al., 2018). However, these factors were not examined directly in the present study and should therefore be regarded as plausible physical interpretations rather than confirmed causal mechanisms.

From a hydrological perspective, declining wet-season rainfall is particularly important because the southwest monsoon supplies most of the annual water input to the province. Persistent reductions during this season may influence reservoir recharge, soil-moisture replenishment, and water availability for agriculture. Although the magnitude of these impacts cannot be inferred from rainfall trends alone, the identified stations represent locations where changes in seasonal rainfall warrant continued monitoring and consideration in water-resource planning.

Table 3.

Wet-season rainfall trends based on the TFPW-MK framework and Sen's slope estimates.

Station	n_{eff}	τ	Z	p	Sen's slope (mm yr ⁻¹)	CI low	CI high	r_i
500001	34	0.062	0.504	0.614	2.90	-8.36	12.95	0.178
500002	34	0.037	0.297	0.767	0.93	-5.77	7.72	0.005
500003	33	-0.261	-2.123	0.034	-7.62	-14.89	-0.59	0.542
500004	34	-0.112	-0.919	0.358	-3.09	-10.82	3.52	0.271
500005	33	-0.258	-2.092	0.037	-5.61	-12.15	-0.13	0.353
500006	33	-0.318	-2.588	0.010	-6.29	-11.51	-1.56	0.428
500007	34	-0.198	-1.631	0.103	-4.41	-9.81	0.68	0.071
500008	34	-0.025	-0.193	0.847	-0.39	-4.62	2.90	-0.218
500009	34	-0.030	-0.237	0.813	-0.54	-4.27	3.06	-0.004
500201	34	0.109	0.890	0.374	3.06	-4.54	9.46	-0.173
500202	33	-0.008	-0.047	0.963	-0.17	-7.89	7.40	-0.363
500301	34	0.027	0.208	0.836	0.73	-6.52	6.38	-0.219

Note: n_{eff} denotes the effective sample size used in the final MK test. Wet-season series requiring pre-whitening had $n_{eff} = 33$ because lag-1 pre-whitening reduces the effective series length by one observation; all other series retained $n_{eff} = 34$

3.5 Dry-season rainfall trends

Dry-season rainfall exhibited predominantly increasing trends across the monitoring network (Table 4; Figure 2). Statistically significant increases were detected at stations 500002, 500004, and 500006, with Sen's slopes of +5.77, +5.22, and +6.78 mm yr⁻¹, respectively (95% confidence intervals excluding zero). No station exhibited a statistically significant decreasing trend.

Although most of the remaining stations also showed positive Sen's slope estimates, their 95% confidence intervals encompassed zero, indicating that the observed increases could not be distinguished from natural interannual variability with statistical confidence. Nevertheless, the predominance of positive slopes suggests an overall tendency toward increasing dry-season rainfall during the study period.

Unlike the annual and wet-season rainfall series, none of the dry-season series exhibited significant lag-1 autocorrelation. Accordingly, dry-season trends were evaluated using the original Mann–Kendall test without pre-whitening, suggesting that serial persistence had a negligible influence on trend detection during the dry season.

From a water-resources perspective, increasing dry-season rainfall may contribute to improved soil-moisture availability and reduced irrigation demand in some areas. However, these increases should not be interpreted as compensating for declining wet-season rainfall because seasonal water availability depends on both the timing and magnitude of precipitation. Together with the wet-season results, these findings indicate that rainfall change in Prachuap Khiri Khan was characterised primarily by contrasting seasonal behaviour rather than by uniform changes in annual rainfall.

Table 4.

Dry-season rainfall trends based on the TFPW-MK framework and Sen's slope estimates.

Station	n_{eff}	τ	Z	p	Sen's slope (mm yr ⁻¹)	CI low	CI high	r_t
500001	33	0.193	1.565	0.118	2.34	-0.97	5.29	-0.087
500002	33	0.341	2.774	0.006	5.77	1.93	10.16	0.153
500003	33	0.030	0.232	0.816	0.51	-3.06	4.42	-0.132
500004	33	0.265	2.154	0.031	5.22	0.57	9.10	0.204
500005	33	0.110	0.883	0.377	2.65	-2.45	7.43	0.142
500006	33	0.379	3.083	0.002	6.78	3.33	10.52	0.068
500007	33	0.117	0.945	0.345	2.56	-1.52	7.34	0.021
500008	33	0.019	0.139	0.889	0.32	-2.42	3.31	-0.231
500009	33	0.110	0.883	0.377	1.52	-0.85	4.64	-0.265
500201	33	0.053	0.418	0.676	1.06	-5.16	8.56	0.044
500202	33	-0.099	-0.79	0.429	-1.07	-7.10	4.16	-0.048
500301	33	-0.042	-0.325	0.745	-0.61	-5.13	4.14	-0.220

Note: n_{eff} denotes the number of complete dry-season observations. All series comprised 33 complete November–April seasons; no series exhibited significant lag-1 autocorrelation and therefore no pre-whitening was applied.

3.6 Evidence of seasonal rainfall redistribution

The most important finding of this study is that rainfall change in Prachuap Khiri Khan Province was expressed more clearly through seasonal variation than through annual rainfall totals. While only one station exhibited a significant annual trend, six significant trends were identified in the seasonal analyses, comprising three wet-season decreases and three dry-season increases. This contrast indicates that annual rainfall alone may underestimate hydroclimatically important changes occurring within individual seasons.

Station 500006 provides the clearest evidence of this pattern. Wet-season rainfall decreased significantly (-6.29 mm yr^{-1}), whereas dry-season rainfall increased significantly ($+6.78 \text{ mm yr}^{-1}$), despite the absence of a significant annual trend (Figure 3). These opposing seasonal trends substantially offset one another when aggregated annually, demonstrating how stable annual rainfall can conceal significant changes in seasonal rainfall behaviour. The term seasonal rainfall redistribution is therefore used to describe contrasting long-term changes in seasonal rainfall while annual totals remain comparatively stable, rather than implying a direct transfer of rainfall from one season to another.

These findings highlight the importance of examining rainfall at the seasonal rather than solely the annual scale. Changes in rainfall timing can influence reservoir recharge, irrigation scheduling, soil-moisture replenishment, and seasonal water availability even where annual rainfall exhibits little long-term change (Azam et al., 2018). Integrating seasonal and annual assessments therefore provides a more comprehensive understanding of hydroclimatic variability and better informs water-resource

Significant Station Trends (TFPW-MK, $\alpha=0.05$) in Prachuap Khiri Khan Province

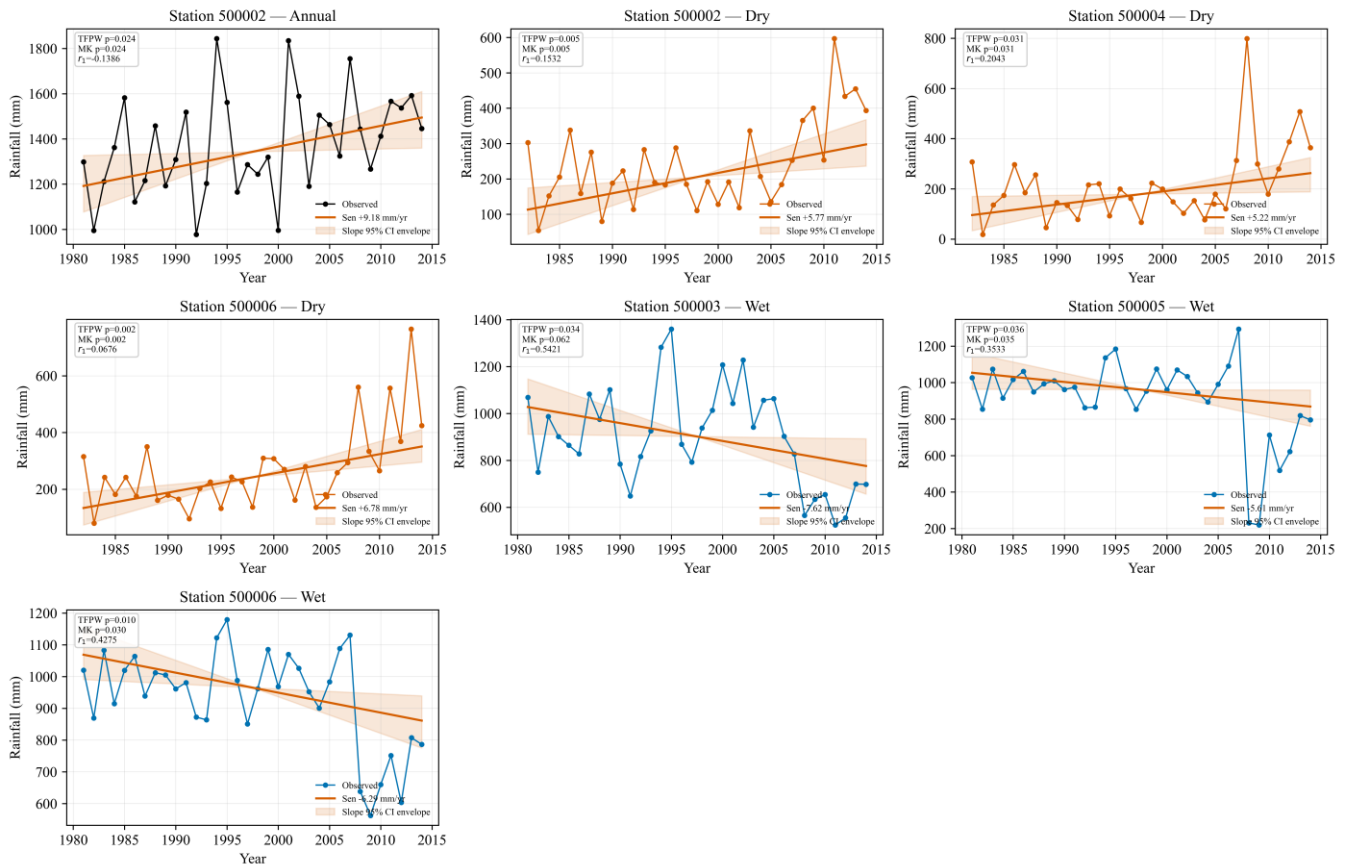


Figure 3. Rainfall time series, Sen's trend lines, and rank-based 95% confidence bounds for the seven station–timescale series with significant TFPW-MK trends (1981–2014).

3.7 Influence of TFPW on statistical inference

Comparison between the original Mann–Kendall (MK) test and the conditional TFPW–MK procedure (Table 5) indicates that the seven significant trends can be grouped according to the influence of serial correlation. Four series (500002 annual, 500002 dry, 500004 dry, and 500006 dry) showed no significant lag-1 autocorrelation; consequently, pre-whitening was unnecessary and both methods produced identical results. Two autocorrelated wet-season series (500005 and 500006) remained significant after pre-whitening, indicating that the detected trends were robust to autocorrelation correction. Only one series (500003 wet) changed from non-significant under the original MK test ($p = 0.062$) to significant after TFPW–MK ($p = 0.034$), suggesting that its statistical significance depends on the treatment of serial correlation. Accordingly, this result should be interpreted with caution, whereas the contrasting seasonal pattern at station 500006 is supported by both methods and therefore provides the strongest evidence for seasonal rainfall redistribution.

Table 5

Comparison of conditional TFPW-MK and original MK results for rainfall series with significant TFPW-MK trends.

Station	Scale	TFPW-MK (Z / p)	Original MK (Z / p)	Interpretation
500002	Annual	2.253 / 0.024	2.253 / 0.024	Identical (no AR)
500002	Dry season	2.774 / 0.006	2.774 / 0.006	Identical (no AR)
500003	Wet season	-2.123 / 0.034	-1.868 / 0.062	Inference changed (tentative)
500004	Dry season	2.154 / 0.031	2.154 / 0.031	Identical (no AR)
500005	Wet season	-2.092 / 0.037	-2.105 / 0.035	Robust after PW
500006	Wet season	-2.588 / 0.010	-2.164 / 0.030	Robust after PW
500006	Dry season	3.083 / 0.002	3.083 / 0.002	Identical (no AR)

Note: Z , standardized Mann–Kendall statistic; p , two-sided p value ($\alpha = 0.05$). Identical (no AR): no significant lag-1 autocorrelation. Robust after PW: remained significant after pre-whitening. Inference changed (tentative): significance changed after pre-whitening.

The only discrepancy occurred for the wet-season rainfall series at station 500003 (Figure 4). The original MK test identified a non-significant decreasing trend ($Z = -1.868$, $p = 0.062$), whereas the conditional TFPW-MK procedure detected a significant decline ($Z = -2.123$, $p = 0.034$). This series also exhibited the strongest positive lag-1 autocorrelation among the wet-season records ($r_1 = 0.542$), exceeding the predefined threshold for pre-whitening. Despite the change in statistical significance, the estimated Sen's slope changed only marginally (-7.77 to -7.62 mm yr⁻¹), indicating that TFPW influenced statistical inference rather than trend magnitude.

Methodological Contrast for Station 500003 (Wet season)

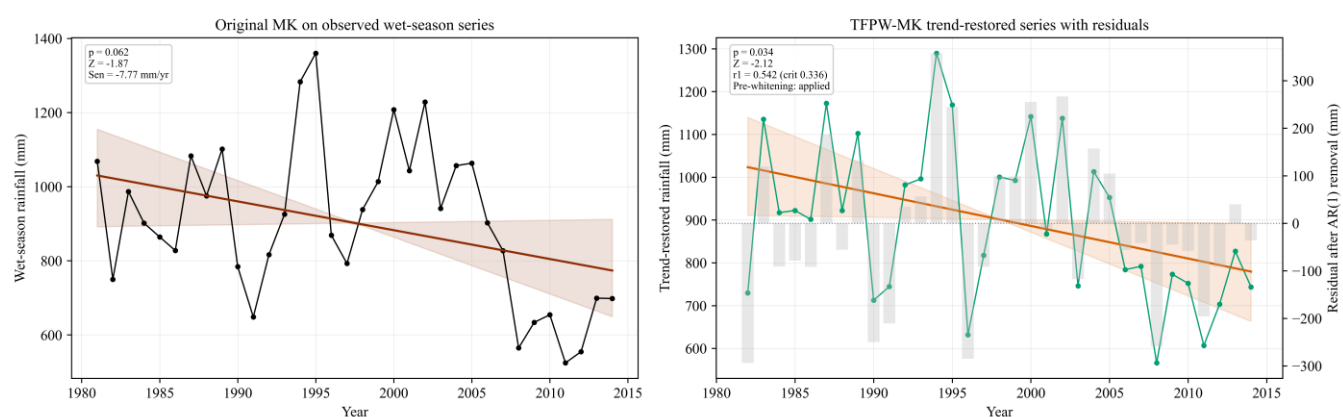


Figure 4. Methodological comparison between original MK and conditional TFPW-MK tests for wet-season rainfall at station 500003.

Overall, the original MK and conditional TFPW–MK procedures yielded consistent results except for one wet-season series (station 500003), whose significance depended on autocorrelation treatment. All other significant trends were either unaffected by serial correlation or remained significant after correction, supporting the use of conditional TFPW for rainfall trend analysis in coastal western Thailand.

3.8 Geographical contrasts in annual and seasonal rainfall trends

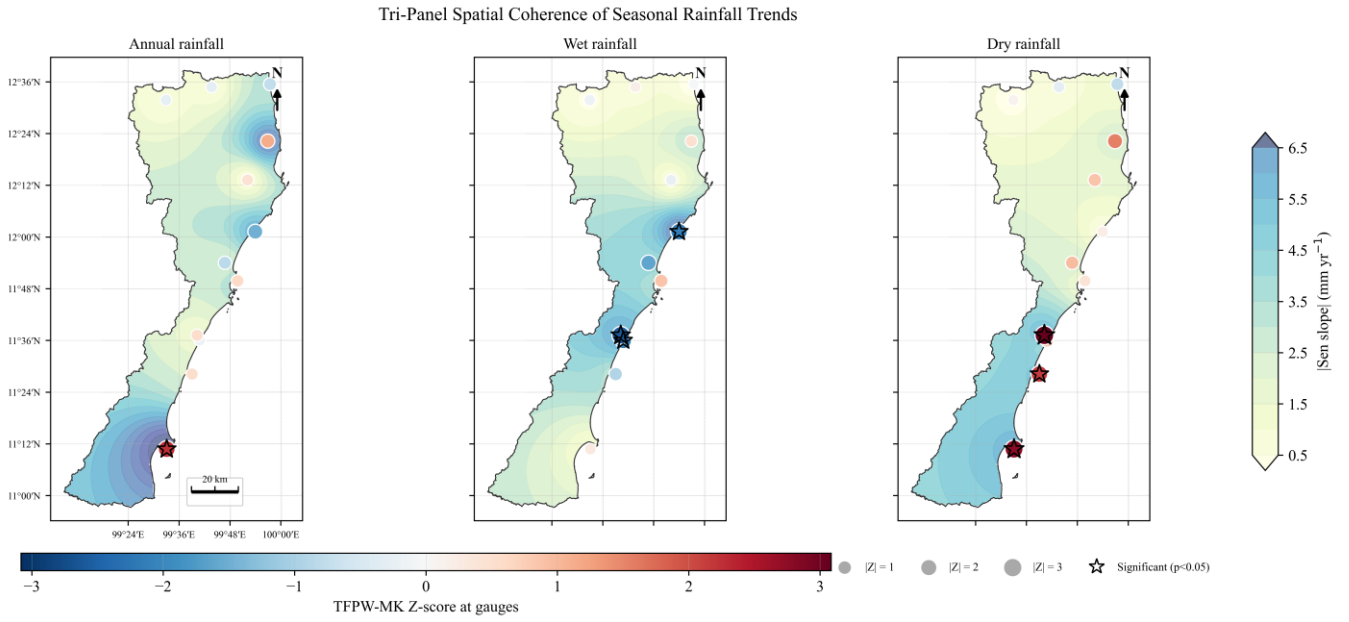


Figure 5. Spatial coherence of annual, wet-season, and dry-season rainfall trends derived from the conditional TFPW-MK analysis in Prachuap Khiri Khan Province (1981–2014).

Figure 5 illustrates the spatial distribution of annual and seasonal rainfall trends by combining station-level TFPW-MK results with interpolated Sen's slope magnitudes. The interpolated background highlights broad geographical patterns in trend magnitude, whereas station symbols represent the direction and strength of the TFPW-MK test, with star symbols indicating statistically significant trends.

The annual rainfall map shows weak spatial organization, with only station 500002 exhibiting a significant increasing trend. Most stations displayed small positive or negative trend magnitudes without statistical significance, indicating that annual rainfall remained generally stable across the province.

In contrast, the seasonal maps reveal more distinct spatial patterns. During the wet season (Wet rainfall map), significant decreasing trends were concentrated at stations 500003, 500005, and 500006, whereas significant dry-season increases were detected at stations 500002, 500004, and 500006 (Dry rainfall map). Station 500006 exhibited significant trends in opposite directions between seasons while showing no significant annual trend, providing the clearest evidence of seasonal rainfall redistribution.

Overall, the spatial distribution of significant trends indicates that rainfall change was localized rather than province-wide. These geographical contrasts demonstrate that seasonal analyses provide

substantially greater insight into hydroclimatic variability than annual rainfall alone and highlight the importance of station-specific information for water-resource planning in coastal western Thailand.

4. Conclusions

This study showed that rainfall change in Prachuap Khiri Khan Province during 1981–2014 was expressed more clearly through seasonal variation than through changes in annual rainfall totals. Annual rainfall remained largely stable across the province, whereas significant wet-season decreases and dry-season increases revealed contrasting seasonal behaviour at several stations. These findings demonstrate that annual rainfall alone may underestimate hydroclimatically important changes in rainfall timing and seasonal distribution (Jiang et al., 2025; Suo et al., 2025).

The clearest evidence of this pattern was observed at station 500006, where significant wet-season and dry-season trends occurred in opposite directions despite the absence of a significant annual trend. This result highlights that seasonal rainfall analysis provides substantially greater insight into hydroclimatic change than annual assessments alone and supports the concept of seasonal rainfall redistribution in coastal western Thailand.

Comparison of the original Mann–Kendall and conditional TFPW-MK procedures showed that the treatment of serial correlation affected the inference for only one series (500003 wet); the other significant trends were either free of significant autocorrelation or remained significant after pre-whitening. These results support a conditional TFPW framework in which pre-whitening is applied only to series exhibiting significant lag-1 autocorrelation.

From a water-resources perspective, the findings emphasize the importance of season-specific and station-specific rainfall assessments (Xie et al., 2025) for irrigation planning, reservoir operation, and climate-change adaptation. Future studies should extend the observational record, incorporate denser monitoring networks, and investigate the climatic drivers responsible for the observed seasonal rainfall changes.

AUTHOR CONTRIBUTIONS

Surasit Punyawansiri contributed to conceptualization, methodology, software development, formal analysis, data curation, visualization, and the preparation of the original draft, as well as reviewing and editing the manuscript.

ACKNOWLEDGEMENTS

The author gratefully acknowledges the Hydro-Informatics Institute (Public Organization) and the Royal Irrigation Department, Thailand, for supplying the rainfall records and supporting analytical resources.

DECLARATION OF CONFLICT OF INTEREST

The author declares no conflict of interest.

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